

Analysis Report

1 Two-Stage Combined Pipeline: Exogenous & Endogenous HR Selection, Fractional Tie-Weighting

- **Pipeline.** Each HR evaluator is paired with an Engineer. HR passes a top- h résumé set to the Engineer, who then hires its top- k from the passed pool. We vary the HR cut four ways: *exogenous* $h = 3$ (17% of 18), $h = 6$ (33%), $h = 9$ (50%), and *endogenous* = the résumés the HR *actually advanced* in the experiment ($\overline{\text{pass}} \approx 58\%$). The $h \times k$ combinations run are: $h = 3: k=1$; $h = 6: k \in \{1, 3\}$; $h = 9: k \in \{1, 3, 5\}$; endogenous $k \in \{1, 2, 3\}$. Arms combine blinding across stages: **Control** (nonblind→nonblind), **Realistic BH** (blind HR → nonblind Eng), **Complete BH** (blind→blind). $k = 1$ is the noisiest cut.
- **Fractional tie-weighting.** At each *score-based* stage the top- k weight is $(k - A)/t$ clamped to $[0, 1]$ (t = score-tie size, A = number strictly above); ties split boundary seats. Compounded: $\text{hire} = f^{HR} \times f_k^{Eng}$. Under endogenous selection the HR pass is the observed binary advance decision ($f^{HR} \in \{0, 1\}$), so only the Engineer stage is fractional; when the HR advances fewer than k , the Engineer hires all of them.
- **IPF matching weights** = $1/(N_{HR} \times N_{Eng})$ within each (arm $\times$ Eng version) block (Complete BH collapsed to one block; weights recomputed within each matched subset in §4).
- **Contrasts & demographic** (label = what the *toggled* stage saw, ref = Blind, sign-flipped so $+$ = blinding increases): *Realistic–Control* = blind HR (uses what HR saw); *Complete–Realistic* = blind Eng (what Eng saw); *Complete–Control* = combined. SEs two-way clustered by HR and Engineer.
- **Caveats.** Significance stars: * $p < .10$, ** $p < .05$, *** $p < .01$ (two-way clustered). Rare groups (esp. Hispanic Female) are noisy in the small pools.

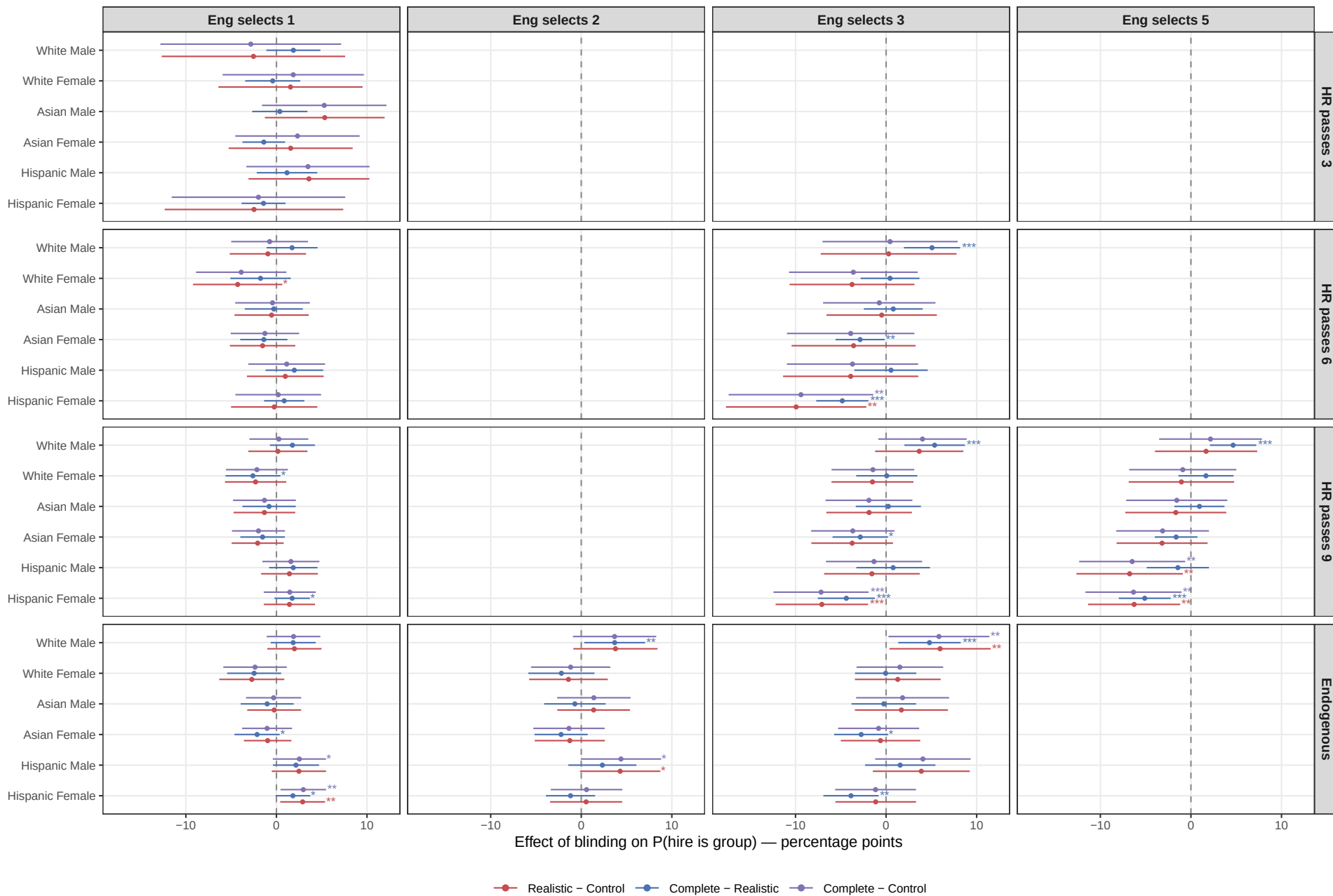
1.1 Treatment Effect on Productivity of Hired (baseline matching)

Effect on true lab productivity (Z) of the hired, SD units; $+$ = blinding raises. Two-way clustered SE; * $p < .10$, ** $p < .05$, *** $p < .01$.

HR passes	Eng selects	Realistic - Control	Complete - Realistic	Complete - Control
3 (17%)	1	-0.025	-0.021	-0.045
6 (33%)	1	-0.018	-0.028	-0.048*
6 (33%)	3	-0.007	-0.024	-0.035
9 (50%)	1	0.004	-0.039	-0.039
9 (50%)	3	0.006	-0.029	-0.030
9 (50%)	5	0.002	-0.029	-0.034
Endog. (~58%)	1	-0.008	-0.038	-0.046
Endog. (~58%)	2	-0.001	-0.028	-0.033
Endog. (~58%)	3	-0.002	-0.026	-0.030

1.2 Treatment Effect on Composition (baseline matching)

Effect of blinding on $P(\text{hire is group})$, percentage points; points = estimate, bars = 95% CI, stars = two-way-clustered significance ($p < .10$, ** $p < .05$, *** $p < .01$). Rows: HR pass; columns: Engineers selected (blank panel = combination not run).*



2 Robustness: How the TE Changes Under Different HR↔Engineer Matching (Eng selects $k = 3$)

Instead of pairing every HR with every Engineer, restrict pairs so HR and Engineer are at firms matching on a characteristic, recompute the IPF weights within the matched set, and re-estimate. **Region** = same U.S. state (evaluator geolocated from the address ZIP, as in the descriptive; 95–96% coverage). Engineer selection fixed at $k = 3$; HR cut shown for exogenous $h = 6, 9$ and endogenous ($h = 3$ is omitted as it runs only $k = 1$).

2.1 Productivity of Hired by Matching Scheme ($k = 3$)

Scheme	HR passes	Realistic - Control	Complete - Realistic	Complete - Control
Baseline (all)	6	-0.007	-0.024	-0.035
Baseline (all)	9	0.006	-0.029	-0.030
Baseline (all)	Endog.	-0.002	-0.026	-0.030
Same firm-size tier	6	-0.007	-0.017	-0.024
Same firm-size tier	9	0.000	-0.036	-0.041
Same firm-size tier	Endog.	0.001	-0.025	-0.028
Same subfield	6	0.006	-0.014	-0.010
Same subfield	9	0.020	-0.019	-0.002
Same subfield	Endog.	-0.010	-0.018	-0.033
Same URM policy	6	-0.004	-0.019	-0.028
Same URM policy	9	0.010	-0.021	-0.015
Same URM policy	Endog.	0.023	-0.021	0.004
Same state	6	0.008	0.000	0.013
Same state	9	0.017	0.005	0.019
Same state	Endog.	0.083*	0.014	0.087

2.2 Composition (Total Effect, Complete – Control) by Matching Scheme ($k = 3$, pp)

Scheme	HR passes	White Male	White Female	Asian Male	Asian Female	Hispanic Male	Hispanic Female
Baseline (all)	6	0.4	-3.6	-0.8	-3.9	-3.7	-9.4**
Baseline (all)	9	4.0	-1.5	-1.9	-3.7	-1.3	-7.2***
Baseline (all)	Endog.	5.8**	1.5	1.8	-0.8	4.1	-1.2
Same firm-size tier	6	-8.1*	-4.8	-3.8	-6.7	-2.4	-8.2*
Same firm-size tier	9	-1.1	-2.0	1.3	-5.7**	0.3	-6.7**
Same firm-size tier	Endog.	2.5	0.9	1.8	-3.8	2.7	-1.3
Same subfield	6	-0.2	-3.6	-2.3	-4.5	-5.3	-10.6**
Same subfield	9	4.3	-2.8	-3.4	-3.4	-3.2	-6.7**
Same subfield	Endog.	8.1***	0.6	1.1	-0.3	4.0	-1.3
Same URM policy	6	-1.8	-1.9	1.1	-3.0	-2.3	-10.6**
Same URM policy	9	2.8	-2.4	-1.2	-2.3	0.3	-5.8*
Same URM policy	Endog.	3.0	0.9	1.0	-0.9	4.8*	0.2
Same state	6	7.1	-10.5*	-9.0	-1.9	-3.3	-5.5
Same state	9	5.7	-3.6	-1.9	-6.8*	-0.3	-4.1
Same state	Endog.	5.5	-4.8	-1.4	-3.5	-2.4	-4.5

$+$ = *blinding raises that group's hire probability.* WM/WF/AM/AF/HM/HF = *White/Asian/Hispanic Male/Female.*